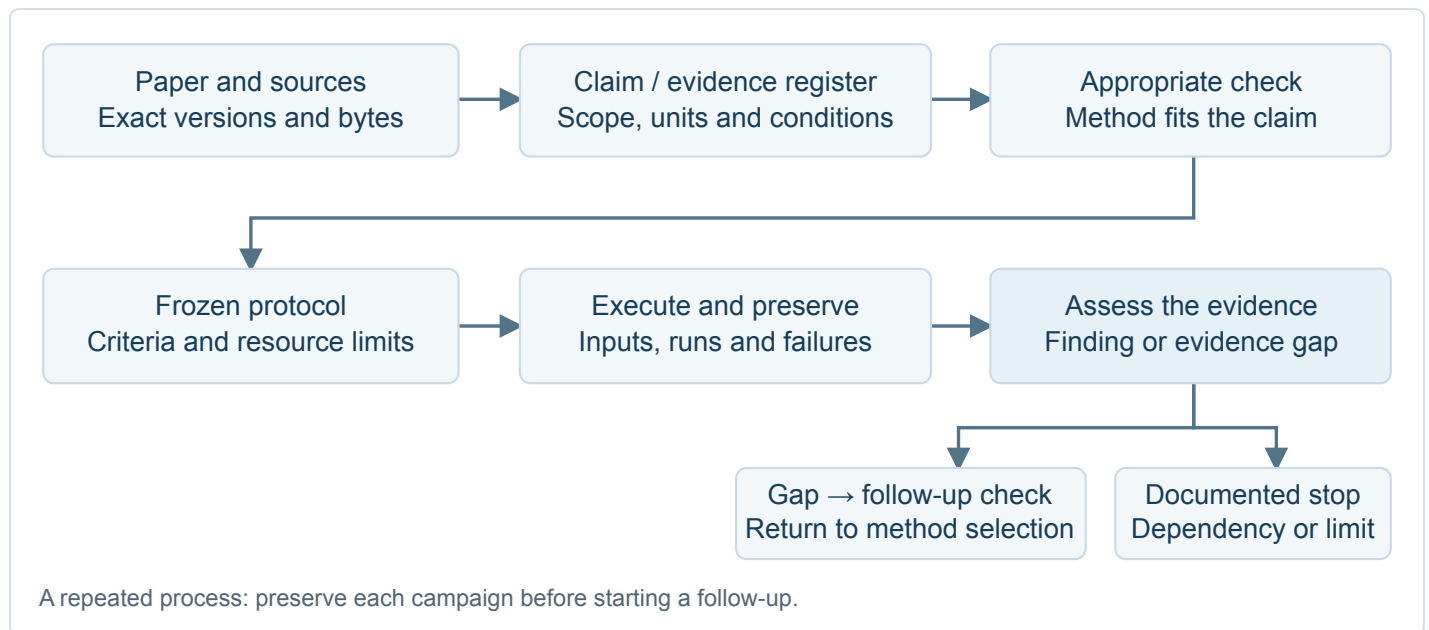


A reusable workflow for verifying research claims

From paper intake to reproducible experiments and targeted follow-ups.

The aim is to produce defensible, claim-by-claim findings under documented conditions. Reducing inconclusive and untested claims means obtaining the evidence each claim needs; additional experiments may strengthen support, reveal a discrepancy or leave a justified limit.



1. Build a precise claim register

Split broad statements into testable claims. Record the source page, table or figure; population; conditions; metric and units; and intended scope. Keep numerical agreement, algorithm correctness, association, causality, generalization and practical effectiveness distinct.

2. Audit the evidence before declaring a dependency missing

Examine supplements, repositories, archived directories, historical APIs and legally available manuscripts. Record retrieval attempts, versions, original-byte hashes and access status. Inventory direct references without recursively expanding their bibliographies.

Track review separately: complete, partial, retrieved but unreviewed, or unavailable. Define complete review and name material not examined. A verified file hash establishes identity, not understanding or review completeness.

Match the method to the claim

No single experimental template fits every paper. Select a method that can address the stated claim and document what it cannot establish. Design choices depend on the question and relevant factors. [1]

CLAIM TYPE	APPROPRIATE CHECK AND ITS LIMITS
Numerical result	Match archived inputs, observation identities, configurations and denominators. Declare units and agreement tolerance before comparison.
Algorithm correctness	Use an independent reference, invariants and boundary cases. Passing finite tests does not prove universal correctness; proof may be required.
Association	Check cohort definitions, identifiers, weighting, stratification and dependence. Descriptive association does not by itself establish causality.
Causal effect	Use an intervention or a justified causal design. State the assumptions needed to identify the effect and which remain untested.
Generalization	Define the target population and test relevant new environments, locations or epochs. Convenience samples limit the scope of inference.
Practical effectiveness	Test end-to-end operation, failures, correctness, performance and resource use. Simulated or isolated component success covers only those conditions.
Theory or standards	Inspect the proof or authoritative primary source. Mark empirical experiments inapplicable when they cannot resolve the claim.

Randomization, blocking and resampling must fit the actual design. Repeated packets, time points or orbital states are not automatically independent replications.

3. Freeze the protocol before each main campaign

- Question, hypotheses, metrics, units and comparison criteria.
- Configurations, input versions, algorithms, dates or epochs and seeds.
- Controls, sensitivity analyses, replication unit and sample-size rationale.
- Exclusions and handling of missing, invalid and ambiguous observations.
- Statistical methods justified by sampling and dependence.
- Resource limits, estimated costs and explicit stopping rules.

Disclose prior exposure to the data. Separate exploratory pilots, main campaigns and follow-ups; retain original protocols, results and amendments. Planned and unplanned analyses should remain distinguishable. [2]

A case reused to tune a method is a development case. It cannot later serve as independent held-out validation.

Preserve evidence and bound conclusions

4. Execute the declared checks

Distinguish reanalysis of recorded data, reproduction with the authors' implementation, independent correctness checks, new empirical measurements and validation of broader claims. Each contributes a different kind of evidence.

Preserve original artifacts and revisions; version corrections separately. Save successful, failed and excluded runs with timestamps, reasons, requested and actual resources, raw bytes, hashes and transformation provenance. Verify stored hashes after retrieval.

Check units, bounds, identifiers, coverage, applicable conservation constraints and algorithm behavior. Compare equivalent configurations and matched observations. Separate published maxima from fixed-budget results.

Independence must be explicit. Implementations sharing inputs can share input errors. Controlled follow-ups distinguish demonstrated discrepancy mechanisms from untested explanations. A released code defect does not establish which revision produced a published result.

5. Assign a finding under the tested conditions

STATUS	MEANING
Reproduced	The relevant check meets the declared criterion under documented conditions. This applies only to the tested scope.
Discrepant	Examined evidence conflicts with the claim or declared criterion. Record comparability limits and unresolved explanations.
Inconclusive	A relevant check or review was completed, but available evidence cannot resolve the claim within the declared scope.
Untested	No adequate check has been completed. Identify the missing check or dependency rather than imply a result.

Numerical agreement, correctness, standards compliance, causality, generalizability and practical effectiveness require separate assessments. Preserve published values separately from reproduced measurements.

6. Turn every unresolved claim into a next-work record

Record the exact gap, evidence or experiment needed, dependencies, feasibility, estimated cost, decision criterion and stopping rule. A small pilot may clarify feasibility; it does not replace the main campaign.

Prioritize central claims and dependencies that unlock several checks. Continue within the authorized scope and resource limits. If a historical dataset, instrument or specialist review remains unavailable, retain that limit and state precisely what the available evidence supports.

Make the process reusable in JumpServe

Share study records, provenance, protocols, campaign execution and public results across papers. Keep scientific implementations specific to each domain.

- 1 **Store linked study records.** Connect sources, atomic claims, protocol versions, configurations, runs, measurements, summaries, artifacts and coverage in Supabase. Preserve missing, invalid, ambiguous and recorded-zero values as distinct states.
- 2 **Add an “Evidence gaps and next experiments” queue.** Each unresolved claim links to its gap record and a proposed campaign. Preserve previous statuses and evidence when new results change an assessment.
- 3 **Expose the evidence alongside the result.** Public pages show tested conditions, sample and coverage counts, exclusions, comparisons, justified uncertainty and downloadable data. Label archived data, new measurements, simulations and model outputs clearly.
- 4 **Keep interpretation and access bounded.** AI can assist extraction, planning and evidence retrieval. Record evaluator and reviewer identities, expertise, human or AI status and independence. Use module-scoped read-only research tools and retain prompt, evidence and tool provenance.

A reusable gap record

FIELD	WHAT TO RECORD
Claim and blocker	Exact claim, current status and the reason evidence is insufficient or the check is unperformed.
Next check	Question, needed inputs, method and scope; identify evidence the check cannot supply.
Execution conditions	Dependencies, replication unit, controls, budget, responsible party and protocol version.
Decision and stop	Criterion for interpreting the result, stopping rule and the limitation retained if blocked.

Report scientific coverage and release readiness separately

A feature can be ready to publish a limited assessment while claims remain unresolved. Declare release-blocking software checks and conditional checks before deployment. Show the scientific limits beside the relevant results; descriptive ranges are not confidence intervals.

These are proposed shared capabilities. This report does not claim that a general-purpose verification engine has been implemented or evaluated.

Supporting methodological sources

- [1] **NIST/SEMATECH e-Handbook of Statistical Methods.** [Choosing an experimental design](#). Guidance on matching designs to objectives and factors; not evidence for a particular paper.
- [2] **Center for Open Science.** [Preregistration](#). Guidance on separating planned and unplanned analyses. Preregistration alone does not establish validity or completeness.